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Studierichting: Informatica
Oefeningenreeks 4A nr. 51.12

$$\begin{aligned}\int \sin^3(x) \cos^2(x) dx &= - \int \cos^2(x) (1 - \cos^2(x)) d(\cos(x)) \\ &= - \int (\cos^2(x) - \cos^4(x)) d(\cos(x))\end{aligned}$$

Stel $t = \cos(x)$

$$\begin{aligned}&= - \int (t^2 - t^4) dt \\ &= -\frac{\cos^3(x)}{3} + \frac{\cos^5(x)}{5} + c\end{aligned}$$

References